drawn, resulting in another set of discriminant functions and corresponding decision boundaries. To put it into other words, the classifier gets a random variable itself. As a consequence, calculating a gradient-based adversarial attack for an input x is done with respect to a drawn set of realizations $\mathcal{A} = \{\theta_1^a, \theta_2^a, \dots, \theta_{S^{\mathcal{A}}}^a\}$, and thus δ from eq. (1) is specific for a realization $f^{\mathcal{A}}(x)$ of the classifier, which we specify by writing $\delta^{\mathcal{A}}$. During inference, the resulting adversarial example $x_{\text{adv}} = x + \delta^{\mathcal{A}}$ is then fed to another random classifier $f^{\mathcal{I}}$, which is based on a different set of realizations from the random vector $\mathcal{I} = \{\theta_1^i, \theta_2^i, \dots, \theta_{S^{\mathcal{I}}}^i\}$. From this perspective, the prediction model $f^{\mathcal{I}}$ is robust against the attack if the distance from x to the decision boundary (given by $f_y^{\mathcal{I}}(x) - f_{c\neq y}^{\mathcal{I}}(x) = 0$) in the *direction of* $\delta^{\mathcal{A}}$ is larger than the length of $\delta^{\mathcal{A}}$, as illustrated in figures 1b) and c).

Robustness conditions for stochastic attacks For a classifier with linear discriminant functions, we are able to turn the previously described observation into a theorem in which we derive a sufficient and necessary condition for the prediction model to be robust against a given attack.

Theorem 4.1 (Sufficient and necessary robustness condition for linear classifiers)**.** *Let* $f : \mathbb{R}^d \times \Omega^h \to \mathbb{R}^k$ *be a stochastic classifier with linear discriminant functions and* $f^{\mathcal{A}}$ *and* $f^{\mathcal{I}}$ *be two MC estimates of the classifier. Let* $x \in \mathbb{R}^d$ *be a data point with label* $y \in \{1, \dots, k\}$ *and* $\arg\max_c f_c^{\mathcal{A}}(x) = \arg\max_c f_c^{\mathcal{I}}(x) = y$*, and let* $x_{adv} = x + \delta^{\mathcal{A}}$ *be an adversarial example computed for solving the minimization problem* (1) *for* $f^{\mathcal{A}}$*. It holds that* $\arg\max_c f_c^{\mathcal{I}}(x + \delta^{\mathcal{A}}) = y$ *if and only if*

$$\min_{c\neq y} \tilde{r}_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2 \ , \textit{ with} \tag{3}$$

$$\tilde{r}_c^{\mathcal{I}} = \begin{cases} \infty \ , & \textit{if} \ \ \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) = \frac{\langle -\nabla_x (f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x)), \delta^{\mathcal{A}} \rangle}{\|\nabla_x (f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2 \cdot \|\delta^{\mathcal{A}}\|_2} \leq 0 \\ \frac{f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x)}{\|\nabla_x (f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}})} \ , & \textit{otherwise} \ , \end{cases}$$

where $\alpha_c^{\mathcal{I},\mathcal{A}}$ *is the angle between* $-\nabla_x(f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))$ *and* $\delta^{\mathcal{A}}$*.*

The proof which is based on Taylor expansion is given in supplement A. The conditions for $\cos(\alpha_c^{\mathcal{I},\mathcal{A}})$ have a nice geometrical interpretation: An angle of more than 90° (which corresponds to a negative cosine value) indicates that the gradient $-\nabla_x(f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))$ and the perturbation $\delta^{\mathcal{A}}$ point into "opposite" directions and thus even for infinitely long moves into the direction of $\delta^{\mathcal{A}}$, the predicted label for x will not change to class $c \neq y$. For positive cosine values, $\tilde{r}_c^{\mathcal{I}}$ specifies the distance to the decision boundary in the attack direction. It looks similar to the minimal distance to the decision boundary in a deterministic setting which is given by $\frac{f_y(x) - f_c(x)}{\|\nabla_x (f_y(x) - f_c(x))\|_2}$ and which is recovered if $\cos(\alpha_c^{\mathcal{I},\mathcal{A}}) = 1$. A cosine value of one however can only occur if the gradient and perturbation direction are identical i.e., if the margin loss is used for calculating the attack direction and if attack and inference model are identical. In practice, the latter is almost surely not the case due to the finite sample approximation, and thus $\cos(\alpha_c^{\mathcal{I},\mathcal{A}}) < 1$. This illustrates the robustness advantage induces by stochasticity.

The derived conditions may locally hold for classifiers which can be reasonably well approximated by a first-order Taylor approximation. To derive further guarantees, we can relax the linearity assumption by assuming discriminant functions which are L-smooth as defined in the following.

Definition 4.1 (L-smoothness [Yang et al., 2022])**.** A differentiable function $f : \mathbb{R}^d \to \mathbb{R}^k$ is L-smooth, if for any $x_1, x_2 \in \mathbb{R}^d$ and any output $c \in \{1, \dots, k\}$:

$$\frac{\|\nabla_{x_1} f_c(x_1) - \nabla_{x_2} f_c(x_2)\|_2}{\|x_1 - x_2\|_2} \leq L \ .$$

For such smooth discriminant function we can derive a sufficient (but not necessary[2]) robustness condition specified by the following theorem, which is proven in supplement A.

Theorem 4.2 (Sufficient condition for the robustness of a L-smooth stochastic classifier)**.** *In the setting of Theorem 4.1, let* $f^{\mathcal{A}}$ *and* $f^{\mathcal{I}}$ *be L-smooth (instead of linear) discriminant functions. Then it*

[2]The necessity of this condition is not given since for non-linear decision boundaries it is possible that behind a region of a different class there exists another region of class y that an attack ends in if $\delta^{\mathcal{A}}$ is long enough.

holds that $\arg\max_c f_c^{\mathcal{I}}(x+\delta^{\mathcal{A}}) = y$ *if* $\min_{c\neq y} r_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ *with*

$$r_c^{\mathcal{I}} = \begin{cases} \infty \ , \ \textit{if} \ \|\nabla_x(f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2} \cdot \|\delta^{\mathcal{A}}\|_2 \leq 0 \ , \\ \frac{f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x)}{\|\nabla_x(f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2 \cdot \cos(\alpha_c^{\mathcal{I},\mathcal{A}}) + \frac{L}{2} \cdot \|\delta^{\mathcal{A}}\|_2} \ , \ \textit{otherwise} \end{cases}$$

with $\cos(\alpha_c^{\mathcal{I},\mathcal{A}})$ *as in theorem 4.1.*

Factors influencing the robustness of stochastic classifiers Since in practice neither the parameter set $\mathcal{A}$ nor $\mathcal{I}$ is fixed, the quantity better describing the practical robustness of a stochastic classifier with linear discriminant functions is

$$\mathbb{P}(\min_{c\neq y} \tilde{r}_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2) \ , \tag{4}$$

where $\tilde{r}_c^{\mathcal{I}}$ and $\delta^{\mathcal{A}}$ are random variables. For L-smooth models, replacing $\tilde{r}_c^{\mathcal{I}}$ by $r_c^{\mathcal{I}}$ leads to a lower bound on the robustness. Deriving an analytic expression for this probability is a hard problem. However, based on theorems 4.1 and 4.2 it becomes clear that a larger $\min_{c\neq y} \tilde{r}_c^{\mathcal{I}}$ relates to increasing the probability in eq. (4) and thus to an increased robustness. We note that $\tilde{r}_c^{\mathcal{I}}$ with $c \neq y$ grows with i) larger prediction margins $f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x)$, ii) smaller gradient norms $\|\nabla_x(f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))\|_2$, and iii) larger angles $\alpha_c^{\mathcal{I},\mathcal{A}}$. Larger prediction margins and smaller gradient norms were also found to positively impact the robustness of deterministic networks [e.g. Ross and Doshi-Velez, 2018]. In contrast, the dependency on the angle is unique to the stochastic setting. We therefore focus on the analysis of this factor in the following.

Analyzing the expected angle The angle $\alpha_c^{\mathcal{I},\mathcal{A}}$ depends on both terms $-\nabla_x(f_y^{\mathcal{I}}(x) - f_c^{\mathcal{I}}(x))$ (for which we use the shorthand $-\nabla_x f_{y-c}^{\mathcal{I}}(x)$ in the following) and $\delta^{\mathcal{A}}$. For further analysis, we first rewrite the gradient as

$$-\nabla_x f_{y-c}^{\mathcal{I}}(x) = \nabla_x \left(\frac{1}{S^{\mathcal{I}}} \sum_{s=1}^{S^{\mathcal{I}}} -f_{y-c}(x, \theta_s^i) \right) = \frac{1}{S^{\mathcal{I}}} \sum_{s=1}^{S^{\mathcal{I}}} -\nabla_x f_{y-c}(x, \theta_s^i) \ .$$

Let $\mu = \mathbb{E}_{\Theta}[-\nabla_x f_{y-c}(x, \theta_s^i)]$ and Σ be the covariance of $-\nabla_x f_{y-c}(x, \theta_s^i)$, then it follows from the central limit theorem that for sufficiently many samples $-\nabla_x f_{y-c}^{\mathcal{I}}(x) \sim \mathcal{N}\left(\mu, \frac{\Sigma}{S^{\mathcal{I}}}\right)$. For simplicity let us assume that the attack is based on the margin loss and only one iteration of gradient ascent. In this case, $\delta^{\mathcal{A}} = \frac{\eta}{\|\hat{\delta}\|_2} \cdot \hat{\delta}$, with $\hat{\delta} \sim \mathcal{N}\left(\mu, \frac{\Sigma}{S^{\mathcal{A}}}\right)$. Note, that we can neglect the scaling of the attack vector, since the angle only depends on $\hat{\delta}$. Therefore, estimating the distribution of $\alpha_c^{\mathcal{I},\mathcal{A}}$ corresponds to estimating the distribution of the angle between two independent multivariate Gaussian random vectors with the same mean and (potentially) differently scaled versions of the same covariance.[3] It is known for vectors from normalized standard multivariate Gaussian distributions that the mode of the distribution of the angle between two random vectors is equal to 90° and that with increased dimension the concentration around this mode gets tighter [Cai et al., 2013]. Unfortunately, deriving a closed form expression for the distribution or expectation in the general case is a challenging task and beyond the scope of this paper. However, we conjecture that the expectation of the angle increases proportionally with the variance and anti-proportionally with the norm of the mean. We illustrate the intuition behind this hypothesis in figure 2. To empirically verify the correctness of this hypothesis we conducted an experiment where we estimated the expected angle between two identically distributed 1,000-dimensional Gaussian random vectors for different choices of means and diagonal covariances. More precisely, we sampled the mean and variances uniformly from $[0, t]$, where t increased from zero to ten with step size 0.2, and estimated the expected angle based on 10,000 vector pairs drawn from the resulting distributions. Results are shown in figure 2 on the right side. As hypothesized, the smaller the length of μ and the higher the average variances, the higher the expected angle.

[3] In the case of more complicated attacks the mean will differ as well but we expect the findings of these section to generalize also to this scenario.

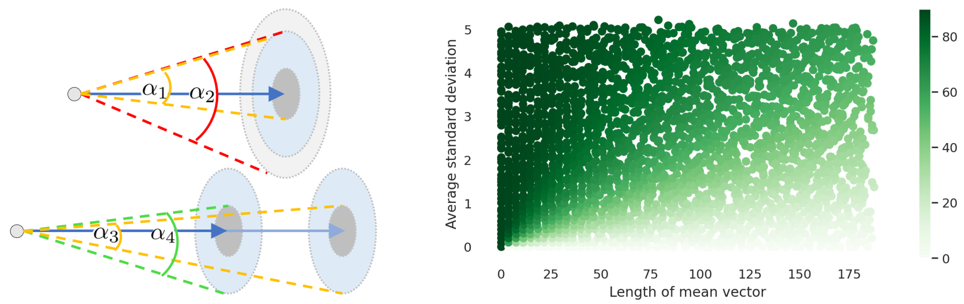

Figure 2: Dependence of the angle w.r.t mean and variance of the gradient. **Left, top**: Sketch of decreased variance which leads to smaller angles. The mean of the gradient is shown as the blue arrow. The circles indicate the areas of high probability for different scales of the covariance matrix. When the covariance is decreased (dark compared to light gray region) the maximum angle between vectors in these region to vectors from the blue region is decreased as well (α_1 compared to α_2). **Left, bottom**: Sketch of increased mean which leads to smaller maximal angles. When the covariance is fixed and the mean is increased (dark compared to light blue arrow) the maximum angle between vectors from high probability regions decreases (α_3 compared to α_4). **Right**: Expected angle between two identically distributed 1,000-dimensional Gaussian random vectors, for Gaussian with different means and variances. Expected angles were calculated based on 10,000 pairwise draws from the respective Gaussian distributions.

Implications for the attack Based on the previous discussion, the only way the attacker can influence the probability of a successful attack, and in this sense the robustness of the model against the attack, is by increasing the amount of samples and thereby reducing the variance of the attack vector. A reduction of the variance leads to a decrease of the expected angle as described in the previous section. This gives a more elaborated explanation of what was often loosely described as finding the "correct" gradient direction in previous work [Athalye et al., 2018]. However, even if the attacker would be able to take infinitely many samples, and thus the variance of the attack direction would be reduced to zero, the expected value of the angle will be larger than zero because of the still existing stochasticity in the inference process. This stochasticity can even lead to an expected angle close to 90° if μ is short, and/or the covariance Σ is high. That is, the advantage of obfuscating the optimal attack direction by incorporating stochasticity into the classifier can be decreased but not fully counterbalanced by taking more samples during the attack. This might explain the finding in He et al. [2019], that the accuracy under attack stagnates at a higher level than the deterministic counterpart when increasing the number of iterations of iterative gradient-based attack methods.

Implications for the model From the perspective of the defender, the analysis of the angle shows that models with an increased gradient variance (which is often associated to a high prediction variance) and a small norm of the mean gradient are connected to larger values of $\alpha_c^{\mathcal{I},\mathcal{A}}$ and thus to a higher probability of unsuccessfully attacks. This explains why including the norm of the mean gradient, the gradient variance, or the angle between gradients as regularization terms in the training of SNNs, as for example proposed by Bender et al. [2020] and Lee et al. [2022][4], lead to an increased empirical robustness.

It would be naturally to suspect that increasing the number of samples used during inference also decreases the robustness, since it decreases the expected angle. However, this is not the case, as it is counterbalanced by an decrease of the norm of the gradient estimate (i.e. the second term in the denominator of $\tilde{r}_c$) with growing sample size.[5] This can be seen by rewriting the expected denominator with respect to the two sample sets $\mathcal{I}$ and $\mathcal{A}$ as

[4]In these works the regularization terms were motivated by maintaining input sensitivity and bounding the expected loss increase, respectively.

[5]We present a preposition showing that the interval incorporating the gradient norm decreases to the norm of μ with increasing amount of samples in supplement A.